

User Control in Tourism Recommender Systems

Group W3

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TUM Uhrenturm

Project Overview & Motivation

- **The Recommender “Black-Box” Problem:** Traditional recommender systems hide their logic, leading users left unknown. [1].
- **Core Goal:** Give users information and control over the ranking logic and filtering mechanisms [4].
- **Regulatory Motivation: EU Digital Services Act (DSA):**
 - *Article 27 (Recommender System Transparency):* Online platforms must explain the main parameters used in recommender systems and provide user-adjustable options [2].
 - *Article 38 (Profiling Options):* Users must be offered at least one option that is not based on profiling [2].

System Architecture & Tech Stack

Fully implemented a working responsive web application:

- **Frontend Framework:** React 18¹ + Vite² for rapid, lightweight development.
- **Mathematical Formula Rendering:** Uses KaTeX³ to render equations directly in browser.
- **Dataset:** Mock database of 20 hotels (10 in Vienna, 10 in Munich) with attributes [4]:
 - *Price per night (USD), Distance from city center (km), Stars rating (1–5), Guest review rating (1–10), and Public transport accessibility score (1–10).*

¹<https://react.dev/>

²<https://vite.dev/>

³<https://katex.org/>

Dual-Algorithm System

The system implements two algorithmic modes that users can toggle [3]:

Algorithm A: Weighted MAUT

- **Approach:** Multi-Attribute Utility Theory (Additive) [5].
- **User Control:** Interactive sliders to define weights (w_i).
- **Output:** 0–100% match score and full ranking of all hotels.

Algorithm B: Constraint-Guided Filtering

- **Approach:** Hard filtering using user-defined bounds.
- **User Control:** Sliders for absolute limit thresholds.
- **Output:** Exact hotels satisfying all limits.

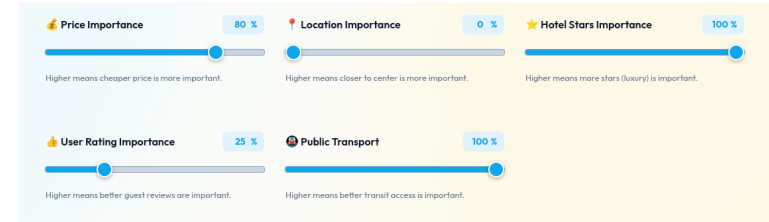


Abbildung: Implemented Sliders for Control

Deep Dive: Weighted MAUT Implementation

Calculates utility $U(X)$ for each hotel X [5]:

$$U(X) = \frac{\sum_{i=1}^5 w_i \cdot u_i(x_i)}{\sum_{i=1}^5 w_i} \times 100\%$$

Walkthrough:

- Attribute Utility ($u_i(x_i)$): hotel's raw values (like a price or distance) are normalized between 0 (worst option) and 1 (best option).
- Weighting (w_i): normalized value is multiplied by slider weight (from 0 to 100) and summed up for all weights and utilities.
- Division ($\sum w_i$): Finally, we divide by the sum of all weights.

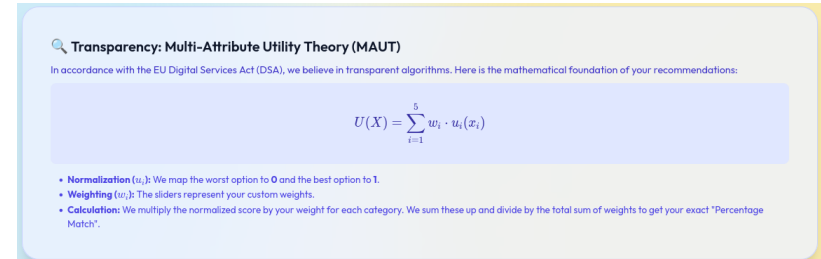


Abbildung: Transparency Banner in UI

Deep Dive: Constraint-Guided Filtering

A hard-limit filtering approach, showing exactly the fitting hotels

- **Pure Logic Over Complex Math:** Instead of calculating utility scores, system checks each hotel against user-defined limits.
- **Inequality Criteria Checked:**

Price $\leq$ Max Price

Distance $\leq$ Max Distance

Stars $\geq$ Min Stars

Rating $\geq$ Min Rating

Public Transport $\geq$ Min Transport

- **Highly Predictable Outcome:** Clear binary classification (a hotel is either included or excluded — no “black-box” scores).
- **Actionable Feedback:** When no hotels match the constraints, the system prompts user to ease their thresholds.

Algorithmic Comparison

Dimension	Weighted MAUT	Constraint-Guided
Core Logic	Utility Maximization (Additive Value)	Hard Boundary Set Reduction
User Interaction	Preference weight sliders (0 – 100%)	Explicit threshold limiters
Outcome	Graduated list (e.g. 92% Match)	Binary selection (Match vs. Out)
Compensability	High (low price compensates location)	Zero (violating one filter excludes hotel)
User Effort	Low (sliders adjust easily)	Medium (requires defining exact limits)
Transparency	Mathematical formula + score breakdown	Simple rule compliance

Tabelle: Algorithmic paradigms for user control.

User Interface & DSA Compliance in Action

Transparency is not just theoretical, but deeply integrated into the user interface:

- **Global Explanation Banner:** Informative card at the top of the dashboard explaining the active algorithm's logic.
- **Interactive Calculations (Local Panel):**
 - Users can expand a “**Show Mathematical Breakdown**” panel on each hotel card.
 - Displays the step-by-step normalized values, weights, points, and final calculation.

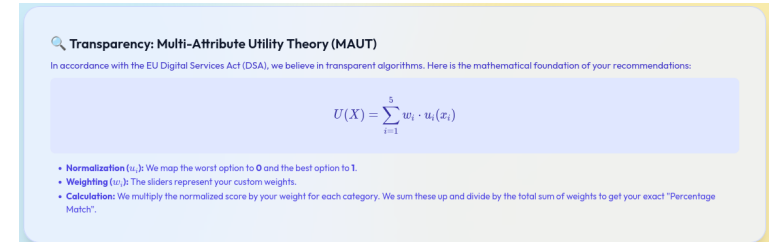


Abbildung: Global Transparency Banner in UI

Detailed Score Breakdown (Full Transparency)

To comply with DSA explainability standards, users can inspect the exact calculations on-demand:

Dynamic Calculation Steps

1. **Feature Normalization:** Converts raw hotel attributes (e.g. \$180 price) to a normalized value $u_i \in [0, 1]$ relative to the current city.
2. **Scaling by Weights:** Multiplies each u_i by the user's custom weight $w_i \in [0, 100]$.
3. **Aggregation:** Sums the individual weighted scores: $\sum w_i \cdot u_i(x_i)$.
4. **Final Division:** Divides the sum by the total active weights ($\sum w_i$) to yield the clean 0 – 100% match score.


Transparency: Multi-Attribute Utility Theory (MAUT)

In accordance with the EU Digital Services Act (DSA), we believe in transparent algorithms. Here is the mathematical foundation of your recommendations:

$$U(X) = \frac{\sum_{i=1}^5 w_i \cdot u_i(x_i)}{\sum_{i=1}^5 w_i}$$

- **Normalization** (u_i): We map the worst option to 0 and the best option to 1.
- **Weighting** (w_i): The sliders represent your custom weights.
- **Calculation:** We multiply the normalized score by your weight for each category. We sum these up and divide by the total sum of weights to get your exact "Percentage Match".

Abbildung: On-Demand Mathematical Breakdown Panel

Realized UI Dashboard

Key UI Elements:

- Clean, premium glassmorphism container design.
- Switchable cities: Munich and Vienna.
- Seamless sliders and dynamic input sync.
- Highlighted match badges with fluid transitions.
- Fully accessible and responsive across mobile and desktop.

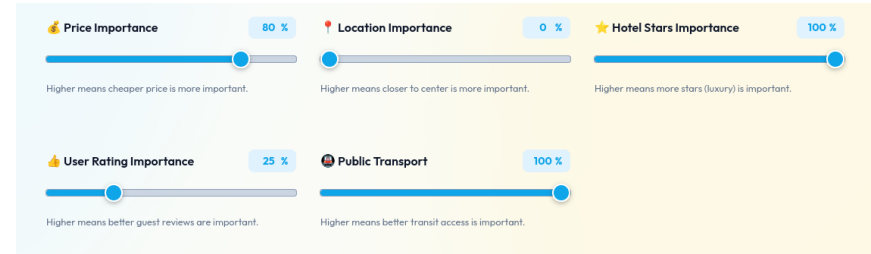


Abbildung: Implemented Web Application Interface

Evaluation & Future Directions

- **Summary of Achievements:**

- Successfully bridged the gap between complex recommendation logic and EU DSA transparency requirements.
- Implemented high-performance, real-time UI rendering of mathematical equations and data filtering.
- Received positive feedback on the click-to-expand mathematical explanation tool.

- **Future Work & Scaling:**

- **Hybrid Recommendation:** Combining filtering with MAUT (e.g. filter out > 300 and then weight remaining).
- **Critiquing Cycles:** Adding critique steps (“Find similar but cheaper”) using utility distance metrics.
- **Dynamic API Integration:** Fetching real-time hotel price and location data.

Thank you for your attention!



Questions?

References I

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- [2] European Parliament and Council of the European Union. *Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act)*. 2022. URL: <http://data.europa.eu/eli/reg/2022/2065/oj>.
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- [5] Christian Schmitt, Dietmar Dengler, Mathias Bauer u. a. „The maut-machine: an adaptive recommender system“. In: *Proceedings of the ABIS Workshop, Hannover, Germany*. 2002.